

Contributors



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capabilities, from predictive maintenance that identifies issues before they occur to personalised driving experiences that adapt to individual preferences. AI also plays a critical role in advanced driver-assistance systems and the gradual progression towards autonomy. Organisations like Waymo and Tesla are leveraging vast datasets to train these systems, pushing the boundaries of what vehicles can perceive and understand. The shift is profound: vehicles are no longer just reactive machines; they are becoming intelligent systems capable of learning and evolving.

EV + SDV: The converging future. The evolution of SDVs is closely intertwined with the rise of electric vehicles. EV architectures, with their simplified mechanical systems and greater reliance on electronics, provide an ideal foundation for software-driven innovation. However, while battery technology remains important, it is no longer the sole differentiator. Increasingly, the competitive edge lies in software capabilities, user experience, and the ability to deliver continuous improvements. In this convergence, energy powers the vehicle, but software defines its identity. The future of mobility will be shaped not just by how far a car can travel, but by how intelligently it can operate.

Beyond technology, SDVs are redefining the economics of the automotive industry. The traditional model, centred on one-off

vehicle sales and periodic servicing, is giving way to a continuous revenue model. Software enables new business models built around subscriptions, feature unlocks, and on-demand services. Automakers can now generate value long after the initial sale, transforming vehicles into ongoing revenue platforms. Companies like General Motors are actively pursuing this shift, while others, such as BMW, are experimenting with subscription-based features. This evolution fundamentally changes the relationship between manufacturers and customers, introducing a more dynamic and service-oriented model. As mentioned by Hariharan Lakshminarayanan, Group Product Manager at Stellantis, “A new ecosystem emerges as vehicles become software-centric; the automotive ecosystem is undergoing a structural transformation. The traditional linear supply chain is giving way to a more interconnected and collaborative model.”

Automakers are increasingly partnering with technology companies, cloud providers, and startups to co-develop solutions. Semiconductor firms, software platforms, and AI companies are now integral to automotive innovation. For instance, Volkswagen Group has made significant investments in building internal software capabilities while also collaborating with external partners. This shift reflects a broader trend towards co-creation,

where innovation emerges from ecosystems rather than isolated entities.

From an electronics engineering perspective, this shift introduces a new set of design challenges. High-performance computing units demand advanced PCB layouts capable of handling high-speed signals, electromagnetic interference, and dense component integration. Thermal management becomes critical, as centralised processors generate significantly more heat than traditional distributed ECUs, requiring efficient heat sinks, liquid cooling, or advanced materials.

Power electronics also take on greater importance, particularly in electric and SDVs, where efficient power distribution directly impacts vehicle range and system reliability. Engineers must balance performance, efficiency, and safety while ensuring robustness across extreme automotive environments. These constraints highlight that while software defines functionality, it remains deeply dependent on sophisticated and reliable electronic design.

With increased connectivity and software dependence come new risks largely absent in traditional vehicles. Cybersecurity threats, system failures, and regulatory challenges have become central concerns. A software vulnerability can impact an entire fleet, amplifying the consequences of any failure. Additionally, questions about liability, particularly regarding autonomous features,